

UNTITLED CAPSTONE PROJECT

GAME DESIGN DOCUMENT

Stealth • Exploration • Atmospheric Survival Horror • Puzzle Adventure

Document status	v0.2 - Working design / pre-production
Working title	TBD (internal placeholder only)
Setting	Napoleonic-era Gothic France
Target playtime	Approximately 4 to 8 hours
Perspective	First-person and third-person prototypes pending
Last updated	2026-07-18

Living-document rule: confirmed decisions are marked LOCKED; unresolved values remain PROVISIONAL, UNDER EVALUATION, OPTIONAL, or TBD until a prototype or playtest resolves them.

NOTE to Professor: *This document was composed with the help of AI, this document will be part of the development process (with AI) going forward, this document was designed under the assumption that the developer (ME) has zero technical knowledge/background in coding or game engine development. My main personal goal for taking this course was to see and learn AI use in an applied professional production environment.*

1. Executive Summary

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The project is a focused, few-hour survival-horror adventure built around exploration, environmental puzzles, scent-driven pursuit, temporary defensive combat, and escape from one interconnected Goth style-themed location.

1.1 Main Pitch

In Napoleonic-era France, the player explores a sealed Gothic mansion or castle (TBD based on project scope and available assets) while a persistent werewolf follows their scent. The player must solve area-based puzzles, manage finite silver ammunition and odor-masking supplies, use safe havens to save progress and craft utilities, and ultimately escape a creature that can be delayed but never permanently killed.

1.2 Game Snapshot

Category	Current Design
Genre	Stealth, walking adventure, atmospheric survival horror, environmental puzzle adventure.
Player fantasy	Survive as a vulnerable but resourceful person who learns the building, reads a predator, and turns scarce historical objects into temporary protection.
Primary objective	Survive - Explore the environment, solve the required puzzle chains, open an escape route, and leave the trapped location.
Failure	Immediate game over when the werewolf physically reaches the player.
Opposing force	One persistent werewolf that searches all active zones and cannot be permanently killed.
Playtime	Approximately 2-4 hours for a first playthrough; final target depends on area count and puzzle density.
World scope	One mansion or castle, organized into 3-5 major areas plus supporting rooms and optional spaces.
Perspective	Final choice open until equivalent first-person and third-person prototypes are compared.
Platform / engine / rating	PC / Unreal / T Rating

1.3 Design Pillars

Pillar	Player-facing promise	Design test
Persistent Predator	The werewolf is an active presence that searches, tracks and re-enters the player's plans rather than waiting only in scripted scenes.	Outside safe havens, the player regularly makes choices based on where the werewolf may be and how strongly it can track them.
Readable Lethality	The creature is deadly, but major attacks have recognizable cues and consistent counterplay.	Players can identify a pounce preparation and explain why a capture occurred.
Scarcity with Agency	Silver, gunpowder and odor supplies are finite, so every use is a strategic decision rather than routine combat.	The game remains finishable after resources are wasted, but the route becomes substantially harder.

Pillar	Player-facing promise	Design test
Atmospheric Discovery	Rooms, objects, sound and architecture communicate the environment, puzzles and history.	Players can orient themselves and infer puzzle logic without constant objective markers.
Temporary Relief	Safe havens and successful neutralizations create short periods of control without eliminating the long-term threat.	Relief is meaningful, limited and followed by renewed uncertainty.

1.4 Scope Boundaries

Ships in the focused game	Deferred, optional or explicitly excluded
One persistent werewolf; one interconnected mansion/castle; 3-5 major areas; area-specific puzzles; scent, sight and hearing; pounce and high-speed charge; hiding; finite silver ammunition; two flintlock weapons; odor masking; 2-3 safe havens; autosave; one complete escape path.	Permanent werewolf kill; enemy roster expansion; conventional action-combat progression; random resource respawns; dodge roll; complex parkour; open world; multiplayer; procedural level generation.
Optional if production supports it: door breaking, werewolf-only shortcuts, castle walls, underground dungeons, and advanced odor interactions.	Any feature that requires substantial new animation, destruction, navigation or content without strengthening the core loop.

2. Player Experience and Game Flow

2.1 Core Loop

Observe the environment and listen for the predator -> choose a route and movement pace -> search for clues, silver, gunpowder, odor supplies or puzzle objects -> solve or advance an environmental puzzle -> evade, hide from, mask against, or temporarily neutralize the werewolf -> reach a safe haven or unlock a new route -> repeat until the final escape condition is completed.

2.2 Player Verbs

Verb	Player-facing result	Risk / cost
Walk	Move quietly at normal pace and preserve stamina.	Leaves a normal scent trail.
Crouch	Move cautiously and reduce visible profile/noise.	Slower traversal; exact scent reduction TBD.
Short sprint	Create immediate distance, evade a pounce sideways, or cross an exposed space.	Consumes stamina; after a threshold, produces a stronger, longer-lasting scent trail.
Contextual vault	Cross selected low obstacles without a parkour system.	Creates movement commitment and likely noise; exact interactions TBD.
Interact / inspect	Open doors, examine clues, operate mechanisms, collect items and use puzzle objects.	May require standing still or expose the player; interruption rules depend on the puzzle.
Hide	Enter authored hiding places and wait for the werewolf to search or leave.	Without active odor masking, the werewolf tracks the scent to the hiding place and pulls the player out.
Apply odor	Use perfume, cologne or absinthe on the player to weaken the scent trail.	Finite consumable; direct sight and nearby sound remain relevant.
Aim / fire	Use the rifle or handgun to stagger or knock the werewolf unconscious.	Consumes one silver ball and one gunpowder charge; every shot is permanent until a save reload.

Verb	Player-facing result	Risk / cost
Reload	Prepare the single-shot firearm for the next defensive use.	Long real-time vulnerability; weapon-specific movement restrictions.
Carry large object	Transport selected puzzle items or containers through the environment.	Occupies the hands and prevents firearm use.

2.3 Session Arc

Phase	Player experience	System focus
Opening / orientation	Learn movement, interaction, basic layout and the immediate escape problem. The werewolf is first suggested, then demonstrated.	Exploration readability; scent premise; first safe-haven discovery.
Early game	Enter the first major area, define avenue in which clues are offered for defensive systems and goal via notes, hints or character dialogue	Hiding, odor masking, first pounce, first firearm use.
Mid-game	Open multiple connected areas, revisit earlier rooms, solve puzzles to progress, and decide when scarce resources justify optional exploration.	Persistent search, stronger scent from sprinting, resource economy, shortcuts and path openings.
Late game	The player understands the environment and predator, but objectives require longer trips through exposed routes.	More frequent line-of-sight encounters, charge pressure, extended puzzle chains.
Final escape	Complete the last environmental condition and survive a final traversal or pursuit toward an exit the werewolf cannot follow. Alternative ending under evaluation TBD	All learned systems; water boundary may become narratively and mechanically relevant. Werewolf cannot swim

2.4 Failure, Saving and Recovery

- **Capture:** Contact or a successful pounce results in immediate failure; there is no conventional health bar currently planned.
- **Saving:** Autosave only, triggered inside true safe havens.
- **Reloading a save:** Restores the complete saved world state, including puzzle progress, resource use and werewolf state recorded at that save.
- **Expedition tension:** Progress since the last safe haven is lost on capture, so spacing must create tension without excessive repetition.
- **Safe-haven reset:** Entering a safe haven ends the active pursuit and resets the werewolf to a randomized patrol state.

3. Player Movement, Stamina and Scent

3.1 Movement Set

- Walking, crouching, short-duration sprinting and contextual vaulting are the complete planned traversal set.
- There is no dodge roll and no complex parkour.
- Player climbing is not part of the current scope unless a specific level requirement proves necessary.
- Large carried objects prevent firearm use and create deliberate vulnerability during transport.

3.2 Stamina-to-Scent Relationship

Stamina is both a movement limiter and a tracking-risk meter. Sprinting consumes stamina. Once expenditure crosses a tuning threshold, the player enters a high-scent state: the trail becomes stronger, may persist longer, and is easier for the werewolf to reacquire. Walking, crouching and resting restore stamina. Exact durations, thresholds, recovery and scent multipliers are UNDER EVALUATION.

Player states	Relative detection	Design purpose
Crouch Stealth / slow movement	Lowest planned baseline; exact reduction TBD. Lower noise radius.	Supports cautious traversal and quieter repositioning.
Walk	Normal continuous scent trail.	Default sustainable exploration pace.
Sprint below threshold	Elevated scent with immediate escape value.	Allows tactical movement without instantly maximizing tracking risk.
Sprint beyond threshold	Strong scent trail with longer persistence and faster reacquisition.	Makes panic running a meaningful long-term cost.
Odor-masked	Trail temporarily weakened or obscured.	Breaks scent pursuit but does not defeat sight or nearby hearing.

3.3 Feedback Requirements

- The player receives readable stamina feedback through breathing, movement performance, animation and/or a restrained interface element.
- Crossing the high scent threshold is not communicated; the player must deduct it.
- The game must distinguish active odor masking, odor expiration, direct visual detection and sound investigation through animation, audio, effects and or HUD support.

4. Werewolf AI and Threat Design

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Smell is the main sensory driver when the player is not in direct line of sight or within hearing range. Sight and sound become decisive only when their local conditions are met.

4.1 Sensory Hierarchy

Sense	Trigger / range	Result	Status
Sight	Direct line of sight.	Confirms the player and supports immediate chase, charge or pounce transitions. Line-of-sight loss returns the system to search/reacquisition logic.	Core rule locked; exact FOV and persistence TBD.
Hearing	Within an authored hearing distance; exact range TBD.	Investigates sprinting, vaulting, doors, dropped objects, firearm discharge and loud puzzle machinery.	Range and source loudness under evaluation.
Smell	Primary sense outside sight/hearing conditions.	Follows the player's scent trail at a slow search/chase pace and locates unmasked hiding places.	Core rule locked; trail model/tuning TBD.

4.2 Core Behaviour Loop

1. The werewolf continuously searches the active environment rather than waiting for the player.
2. When it intersects or reacquires a usable scent trail, it follows that trail at a slow pursuit pace.
3. Direct sight or sufficiently close sound escalates the pursuit and may enable a charge or pounce.
4. When odor masking weakens the trail and no stronger sensory evidence exists, the werewolf transitions to randomized patrol/pathing.
5. After losing direct information, it searches the local area, then resumes patrol until it reacquires the player.

4.3 State Model

State	Typical trigger	Behaviour / exit
Random patrol	No reliable scent, sight or sound target; safe-haven reset; successful odor masking.	Moves through valid zones and selected routes. Exits when a sensory cue is acquired.

State	Typical trigger	Behaviour / exit
Scent pursuit	Usable player scent trail.	Follows the trail at a slow tracking pace. Strong sprint scent improves reacquisition.
Sound investigation	Audible event in hearing range.	Moves toward the source and searches. Exact certainty and duration TBD.
Visual chase	Direct line of sight.	Runs or positions for a charge/pounce according to distance and geometry.
Pounce preparation	Player enters a valid pounce distance/angle.	Lowers profile and visibly commits weight before launching.
Pounce	Telegraph completes.	Leaps toward the player; player can counter by sprinting sideways at the moment of commitment.
Charge	Longer clear pursuit route.	High-speed ground sprint toward the player. Turning, collision and counterplay are under evaluation.
Hiding-place search	Scent leads to a hiding place or local search reaches it.	Unmasked player is found and pulled out. Masked player causes a stalled search animation, after which patrol resumes.
Staggered	Silver body hit.	Short interruption; duration TBD; returns to pursuit/search after recovery.
Unconscious	Silver head hit.	Extended neutralization; duration TBD; eventually recovers and resumes hunting.

4.4 Navigation and Space Rules

- The werewolf can travel between all active zones.
- It passes through open doors.
- It can climb using authored navigation links and animations.
- It can search hiding places finding the player hiding spot results in game over.
- Door breaking is OPTIONAL and depends on destruction assets, animation and navigation feasibility.
- Werewolf-only shortcuts are UNDER EVALUATION. They must not create visible teleportation or unfair instant reappearance.
- True safe havens are inaccessible to the werewolf.

4.5 Pounce Rules

- At a valid distance, the werewolf lowers its profile and displays a recognizable preparation animation.
- The player counters by timing a sideways sprint as the werewolf commits to the leap.
- There is no dodge-roll invulnerability; success comes from reading the animation and moving out of the committed path.
- A missed pounce must create a brief recovery window. Activation distance, telegraph time, trajectory width, midair correction and recovery duration are UNDER EVALUATION.
- The sprint used to evade raises scent risk, linking immediate survival to later tracking pressure.

4.6 Fairness Rules

- The player should be able to identify the sensory cause of a chase or capture.
- Major lethal actions require readable visual and audio cues.
- No normal-pathing teleportation in the player's view.
- Every required route must remain survivable without ammunition or odor supplies, although the no-resource path may be extremely difficult.
- Charge counterplay, hiding edge cases and direct-sight interaction with odor masking must be resolved by prototype before content lock.

5. Defensive Combat, Resources and Crafting

5.1 Combat Philosophy

Combat is temporary neutralization, not victory. The werewolf cannot be permanently killed. Firearms exist to interrupt an attack, create distance, open a short puzzle window or make a dangerous route temporarily safer. Scarcity prevents the game from becoming a shooter.

5.2 Firearm Specification

Property	Flintlock long rifle	Flintlock handgun
Capacity	Single shot.	Single shot.
Ammunition	Shared silver-ball ammunition; each shot also consumes one gunpowder charge.	Same.
Reload	Manual, approximately 12 seconds in real time.	Manual; duration 9 seconds in real time..
Movement during reload	Player cannot move.	Player may move but cannot sprint.
Accuracy	Best range and deliberate headshot potential; accuracy improves while standing still.	More practical in confined spaces; accuracy also improves while standing still.
Misfire	No random misfires.	No random misfires.
Open questions	Reload interruption/stage persistence; aim sway; perspective-specific handling.	Reload duration; aim sway; close-range hit reliability.

5.3 Hit Outcomes

Hit result	Werewolf response	Player opportunity
Body shot	Immediate stagger and attack interruption for a short duration.	Escape one room, break line of sight, enter a route, or finish a brief interaction.
Headshot	Collapse and temporary unconsciousness for a substantially longer duration.	Explore nearby rooms, move a large object, complete a puzzle stage, or reach another zone.
Miss	No defensive benefit; resources are consumed.	Player must continue evasion or reload under the weapon's restrictions.

Head hit volume, aim assistance, sway, pounce/charge shooting and exact neutralization timers remain UNDER EVALUATION. All in-world timers continue during inventory, clue review, puzzle interactions and crafting; no diegetic interface pauses recovery or odor duration.

5.4 Finite Resource Economy

Resource	Source and conversion	Rules
Silver objects	Authored silverware, jewelry and approved decorative items.	Each eligible silver object converts to exactly one silver ball. No respawn.
Gunpowder	Finite authored caches such as powder horns, military storage, armories or secured rooms.	One charge is consumed per shot. Silver without gunpowder is not usable ammunition.
Odor supplies	Perfume, cologne and absinthe placed in authored locations.	Applied to the player only in the current design; temporarily weakens scent. Quantity does not respawn.
Key / puzzle objects	Area-specific discoveries.	Important narrative or puzzle objects cannot be accidentally melted or consumed.

5.5 Odor Masking

- Perfume, cologne and absinthe are applied directly to the player.
- The effect weakens or obscures the scent trail and places the werewolf into randomized patrol/pathing when no sight or sound cue overrides it.
- Odor does not make the player invisible and does not suppress nearby noise.
- When the player hides while masked and the werewolf lacks direct confirmation, it stalls in a search animation and eventually resumes patrol.
- Additional uses – throwing, scent barriers, lures, item-specific strengths, or interaction with sprinting – are UNDER EVALUATION and not required for the first prototype.

5.6 Crafting, Inventory and Safe Havens

System	Locked rule
Crafting location	Crafting occurs inside true safe havens.
Crafting speed	Instantaneous once the player chooses eligible silver objects and has gunpowder.
Inventory	Player carries both firearms, ammunition, gunpowder, a small number of odor consumables, key items and small puzzle objects.
Large objects	Physically carried; occupy the hands and prevent weapon use. Water containers have no general defensive interaction and are included only if a puzzle/endgame requires them.
Resource exhaustion	No resource respawns. The game remains completable after every bullet is missed and every odor item is wasted, but becomes extremely difficult.
Exact quantities	Determined after level layout, encounter count, expected accuracy and optional exploration rewards are known.

6. World, Areas, Puzzles and Pacing

6.1 Environment Choice

The game takes place in Napoleonic-era Gothic France. The final environment is either a mansion or a castle. The choice remains PROVISIONAL because it affects scale, traversal, available exterior spaces, military facilities, walls, water access, asset burden and narrative framing.

6.2 Major-Area Model

The environment contains at least 3 and no more than 5 major production areas. Named rooms are grouped into those areas; not every room becomes a separate zone. Exact adjacency, puzzle ownership and final room list will be iterated during game production.

Provisional major area	Included location pool	Gameplay role / puzzle status
Central / public wing	Main hall, dining hall, concert hall, courtyard or small plaza.	Primary navigation hub; landmarks, route gates, large pounce/charge spaces. Area puzzle TBD.
Residential / private wing	Living quarters, lord's room, guest or servant rooms as needed.	Narrative clues; perfume/cologne and jewelry; tighter corridors and hiding. Area puzzle TBD.
Service / storage wing	Warehouse, kitchens/pantry/workshop if required.	Gunpowder, silverware, crafting support, machinery and large-object transport. Area puzzle TBD.
Military / medical wing	Barracks, medical ward, armory/powder storage if required.	Firearm/resource discovery, evidence of prior resistance, dangerous high-value spaces. Area puzzle TBD.

Provisional major area	Included location pool	Gameplay role / puzzle status
Exterior grounds	Garden, cemetery, courtyard/plaza; castle walls only if castle is selected.	Open charge threat, limited cover, cemetery clues, final-route setup and possible water boundary. Area puzzle TBD.
Optional underground	Underground dungeons, crypts, tunnels or flooded passages.	OPTIONAL. Included only if it delivers a necessary puzzle, shortcut, werewolf origin clue or endgame water access.

6.3 Location Pool

Current approved location pool: dining hall; concert hall; main hall; living quarters; medical ward; warehouse; garden; lord's room/private chamber; cemetery; optional underground dungeons; castle walls if the castle setting is selected; courtyard or small plaza; and barracks. Additional supporting rooms may be added, combined or removed as production and level flow require.

6.4 Provisional Flow Logic

Flow layer	Current rule
Hub	The main hall or central public wing provides recognizable orientation and connects several locked or obstructed routes.
Area progression	Each major area owns at least one puzzle chain or critical objective that changes world access.
Revisiting	New keys, mechanisms, shortcuts or information can send the player back through earlier areas under changed werewolf pressure.
Safe havens	Two or three true safe havens are distributed according to traversal time and failure cost, not simply one per area.
Final route	The last puzzle chain opens an exit or crossing. The werewolf's inability to swim may become the endgame constraint.
Iteration	Room count, adjacency, shortcuts, puzzle order and optional content remain adjustable through greybox testing.

6.5 Puzzle Design Framework

The individual puzzles have not yet been crafted. The GDD therefore locks their role and quality requirements without inventing solutions.

Requirement	Rule
Area ownership	Each major area contains a puzzle chain or objective that justifies entering and learning that space.
Environmental grounding	Puzzle logic must fit the building's function, period objects and architecture rather than appear as an unrelated minigame.
Interruption	Puzzles occur in the active world. Whether partial progress persists and how quickly the player can disengage are determined per puzzle.
Threat integration	The werewolf can interrupt or pressure puzzle work, but required interactions must remain possible without spending finite resources.
Clarity	The environment provides sufficient clues and feedback to understand the goal, valid actions and state changes.
Backtracking	Revisiting is allowed when it produces new information, route mastery or changed danger rather than empty repetition.
No softlocks	Puzzle-critical and narrative-critical items cannot be consumed as ammunition or lost permanently.

6.6 Pacing and Safe-Haven Placement

- Quiet exploration and clue interpretation alternate with search pressure, direct chase, resource decisions and short recovery windows.
- Two or three true safe havens support autosaving, clue review and crafting.
- Entering a safe haven resets the werewolf to random patrol. Exit placement and patrol spawning must prevent immediate unavoidable capture or safe-room door exploitation.
- Exact travel time between saves, neutralization usefulness and resource placement depend on the final greybox and puzzle order.

7. Narrative, Setting and Atmosphere

7.1 Locked Setting Direction

- Napoleonic-era France with a Gothic mansion or castle as the primary environment.
- Period-compatible flintlock firearms, loose gunpowder, silverware, jewelry, perfume, cologne and absinthe support both fiction and mechanics.
- The architecture should plausibly contain residential, public, service, military, medical, exterior and burial spaces selected from the approved location pool.

7.2 Narrative Framework - To Be Crafted

TBD

The protagonist, reason for entering or becoming trapped, identity/origin of the werewolf, purpose of the major puzzles, safe-haven fiction, and final outcome have not been authored yet.

Narrative question	Required design outcome
Who is the protagonist?	Explains period-appropriate firearm handling, motivation, knowledge and emotional stakes without demanding excessive exposition.
Why can the player not leave?	Creates the initial escape problem and gives the major puzzles a coherent shared purpose.
Why is the werewolf trapped here?	Supports its access to all zones and its inability to leave or cross water.
Why are safe havens protected?	Provides a consistent reason the werewolf cannot enter and avoids arbitrary game-only rooms.
What do the puzzles accomplish?	Each area objective contributes to opening, powering, revealing or reaching the final escape route.
What does escape mean?	Resolves the mechanical objective and clarifies whether the werewolf remains contained, follows, or is separated by water.

7.3 Atmosphere and Environmental Storytelling

Layer	Direction
Lighting	Low-light Gothic mood with readable navigation, interactables and attack telegraphs. Darkness cannot be the sole source of difficulty.
Soundscape	Directional movement, breathing, claws, doors, distant activity and intentional silence communicate location and threat state.
Werewolf presentation	Partial sightings, silhouette, posture and sound preserve mystery while maintaining readable pounce, charge, stagger and recovery cues.
Environmental storytelling	Room condition, abandoned preparations, consumed resources, barricades, medical evidence and personal possessions suggest prior events and puzzle logic.

Layer	Direction
Music	Restrained; supports major escalation and relief without replacing spatial audio used for gameplay.

8. UX, UI and Accessibility

8.1 Camera Decision Prototype

The final camera is intentionally open. The team will build the same compact sequence in first person and third person: one exploration section, one pounce encounter and one firearm encounter. The comparison will evaluate environmental inspection, motion comfort, hiding readability, pounce telegraph recognition, headshot aiming, spatial awareness, animation burden and overall horror tone.

8.2 Baseline Information Requirements

- Minimal HUD that still communicates stamina/high-scent risk, loaded/unloaded firearm state, ammunition, gunpowder and active odor masking.
- Clear interaction prompts and state feedback for locked, available, collected, crafted, solved, autosaved and failed states.
- Clue review available inside safe havens at minimum; whether it is available everywhere remains under evaluation.
- No in-world inventory, clue or puzzle interface pauses werewolf recovery, odor or pursuit timers.

8.3 Control Baseline

Action group	Required behavior
Movement	Walk, crouch, short sprint and contextual vault; no dodge roll.
Interaction	Contextual inspect/use/pick up; large-object carry state blocks weapon use.
Weapons	Aim, fire and manual reload; rifle locks movement for about 12 seconds; handgun allows walking but not sprinting.
Inventory / odor	Apply finite odor consumable to player; show effect state and expiry feedback.
Pause / options	Menu behavior and whether true system pause halts simulation are technical/accessibility decisions to resolve.

8.4 Accessibility and Difficulty - Under Evaluation

Area	Prototype / design question
Controls	Full remapping, hold/toggle options, aim sensitivity and alternatives for repeated/long holds.
Visual readability	Subtitle/text scaling, contrast, brightness, non-color state cues and reduced camera shake.
Hearing	Captions and optional visual equivalents for critical directional werewolf cues without removing uncertainty.
Pounce timing	Optional wider telegraph/evasion window that preserves the core sideways-sprint counter.
Aiming	Optional aim assistance or generous head hit volume without turning firearms into reliable offense.
Puzzle support	Layered hints and repeatable clue review once puzzle content exists.
Difficulty	Adjust detection forgiveness, resource abundance, save spacing or telegraph windows rather than simply increasing werewolf speed.

9. Technical Strategy and Production Plan

9.1 Required Technical Systems

System	Minimum requirement	Main risk / validation
Perspective prototype	Equivalent first- and third-person test spaces with the same inputs and encounter conditions.	Avoid evaluating different content rather than the camera itself.
Werewolf AI	State machine or behavior tree covering patrol, scent pursuit, hearing investigation, sight chase, pounce, charge, hiding search, stagger and unconsciousness.	Fair transitions and cross-zone navigation.
Scent trail	Player trail with activity-based strength, persistence and odor masking.	Performance, debugging and player comprehension.
Navigation	All-zone navigation, open-door traversal, authored climbing links; optional door destruction and shortcuts.	Getting stuck, unfair shortcuts, animation/nav mismatch.
Combat	Single-shot weapons, reload states, hit-region responses and persistent timers.	Camera-specific aiming and interruption edge cases.
Inventory/crafting	Finite authored items, protected critical objects, one-object-to-one-ball conversion and gunpowder charges.	State persistence and softlock prevention.
Save system	Autosave only in safe havens; restore full world/resource/puzzle state.	Excessive replay distance and inconsistent AI reset.
Puzzle framework	Stateful in-world interactions with per-puzzle interruption/persistence rules.	Threat pressure making puzzle reasoning impossible.

9.2 Production Milestones

Milestone	Playable result	Exit criteria
Risk prototype	Movement, stamina/high-scent threshold, scent trail, odor masking, one hiding place, one pounce, one firearm and one safe haven.	Core states are debuggable; pounce can be read and evaded; scent/odor loop is understandable.
Perspective comparison	Same exploration, pounce and firearm encounter in first and third person.	Camera choice recorded with evidence.
Greybox loop	A hub and two connected areas, one placeholder puzzle chain, cross-zone werewolf navigation and autosave.	10-20 minute loop works without scripting around AI failures.
Vertical slice	One representative major area with final target quality for art, audio, puzzle, threat, resource and save flow.	Target experience is playable and production cost is measured.
Content complete	All selected major areas, puzzles, safe havens and ending playable start-to-finish.	No softlocks; no-resource completion path exists.
Polish / release candidate	Tuned encounter fairness, accessibility baseline, performance, audio mix and final save spacing.	Critical issues closed and 2-4 hour target validated.

9.3 Constraints and Risks

Risk / constraint	Why it matters	Mitigation / fallback
Persistent predator fatigue	Continuous pressure can prevent exploration and puzzle reasoning.	Use safe havens, search pacing, loss-of-trail states, authored quiet windows and area-specific encounter tuning.

Risk / constraint	Why it matters	Mitigation / fallback
AI navigation scope	All-zone travel, climbing, doors and shortcuts are expensive and failure-prone.	Prioritize open-door traversal and authored climb links; cut door destruction/shortcuts if unstable.
Animation burden	Pounce, charge, climbing, pulling from hiding, weapon states and two camera perspectives require significant animation.	Prototype with blockout animation; finalize one perspective before content production; treat door breaking as optional.
Finite-resource balance	Too few resources feels punitive; too many turns temporary defense into routine combat.	Place authored resources after greybox; test completion with average use, wasteful use and zero remaining supplies.
Autosave repetition	Long distances between safe havens can make immediate capture frustrating.	Measure traversal and puzzle replay time; add or reposition havens before art lock.
Puzzle-threat conflict	Unpaused predator pressure can make complex puzzles unreadable.	Design interruptible stages, preserve logical progress, provide local escape routes and tune patrol timing per area.
Mansion/castle scope	Too many named rooms can expand the project beyond a focused game.	Group rooms into 3-5 major areas; cut or combine any room without a puzzle, progression, narrative or encounter purpose.

9.4 AI-Assisted Development and Tooling

The game-development AI workflow, named agents/tools, token budget, engine, version-control platform and hardware targets are not yet selected. This section remains a required production follow-up rather than inventing commitments. At minimum, any AI-assisted workflow must define concrete outputs, human approval gates, privacy/IP boundaries and a plausible usage budget before implementation begins. Given my very limited Unreal Engine knowledge and minimal budget restrictions every decision taken must support best output with minimal resources while maximizing AI assistance which was the main driver for taking this course.

10. Prototype Plan, Open Questions and Decision Log

10.1 Highest-Priority Prototype Questions

Priority	Question	Test / evidence
1	First person or third person?	Build the same exploration, pounce and firearm sequence in both; compare readability, aiming, animation cost and horror tone.
2	What pounce values feel lethal but fair?	Tune activation distance, telegraph, sideways-sprint window, leap width and miss recovery with repeated player tests.
3	How should scent be represented?	Debug trail strength/persistence; test whether players understand normal, high-scent and odor-masked states.
4	What are valid sight and hearing ranges?	Test representative rooms/corridors and verify the player can explain investigation/chase causes.
5	How long should body stagger, head unconsciousness and odor masking last?	Measure the meaningful actions each window allows in the greybox rather than choosing abstract seconds.
6	How should charge work?	Prototype turning, acceleration, obstacle collision and no-resource counterplay.
7	How far apart should safe havens be?	Measure replay time, tension and resource/crafting cadence in the final area flow.

Priority	Question	Test / evidence
8	Which areas and puzzles ship?	Greybox 3-5 major areas; keep only spaces with a clear puzzle, progression, encounter, resource or narrative purpose.

10.2 Open Questions

Topic	Current status
Perspective	UNDER EVALUATION through matched prototype.
Mansion or castle	PROVISIONAL; decide after area layout, narrative and asset-cost comparison.
Puzzles	TO BE CRAFTED; one chain per major area is the current structural requirement.
Narrative and ending	TO BE CRAFTED.
Charge rules	UNDER EVALUATION.
Headshot/aim model	UNDER EVALUATION.
Handgun reload time	TBD.
All timers	Durations TBD; continue during all in-world interactions.
Door breaking	OPTIONAL based on technical/asset cost.
Werewolf shortcuts	UNDER EVALUATION.
Difficulty/accessibility	UNDER EVALUATION after core prototype.
Platform, engine, team, schedule and rating	TBD.

10.3 Decision Register

Status	Decision
LOCKED	2-4 hour focused stealth/puzzle survival-horror experience in Napoleonic-era Gothic France.
LOCKED	Walking, crouching, short sprint and contextual vault; no dodge roll or complex parkour.
LOCKED	Smell is the primary tracking sense outside direct sight/hearing conditions; sprinting past a stamina threshold strengthens scent.
LOCKED	Odor is applied to the player and temporarily breaks scent pursuit into random patrol when not overridden by sight/sound.
LOCKED	Werewolf crosses all zones, open doors, climbs and searches hiding places; safe havens are protected.
LOCKED	Pounce is telegraphed by lowered profile and countered by a timed sideways sprint.
LOCKED	Flintlock rifle and handgun are single-shot, share silver-ball ammunition, consume gunpowder and only temporarily neutralize the werewolf.
LOCKED	Rifle reload is approximately 12 seconds with no movement; handgun permits walking but not sprinting during reload.
LOCKED	Each eligible silver object becomes one bullet; crafting is instant in safe havens; finite resources never respawn.

Status	Decision
LOCKED	Autosave only in 2-3 safe havens; entering one resets the werewolf to random patrol.
LOCKED	Werewolf cannot swim; water has no general interaction but may become relevant to the endgame.
PROVISIONAL	Mansion/castle comprises 3-5 major areas assembled from the approved location pool; final layout evolves during production.

END OF CURRENT DESIGN BASELINE - UPDATE THIS DOCUMENT WHEN A PROTOTYPE OR PLAYTEST CHANGES A RULE